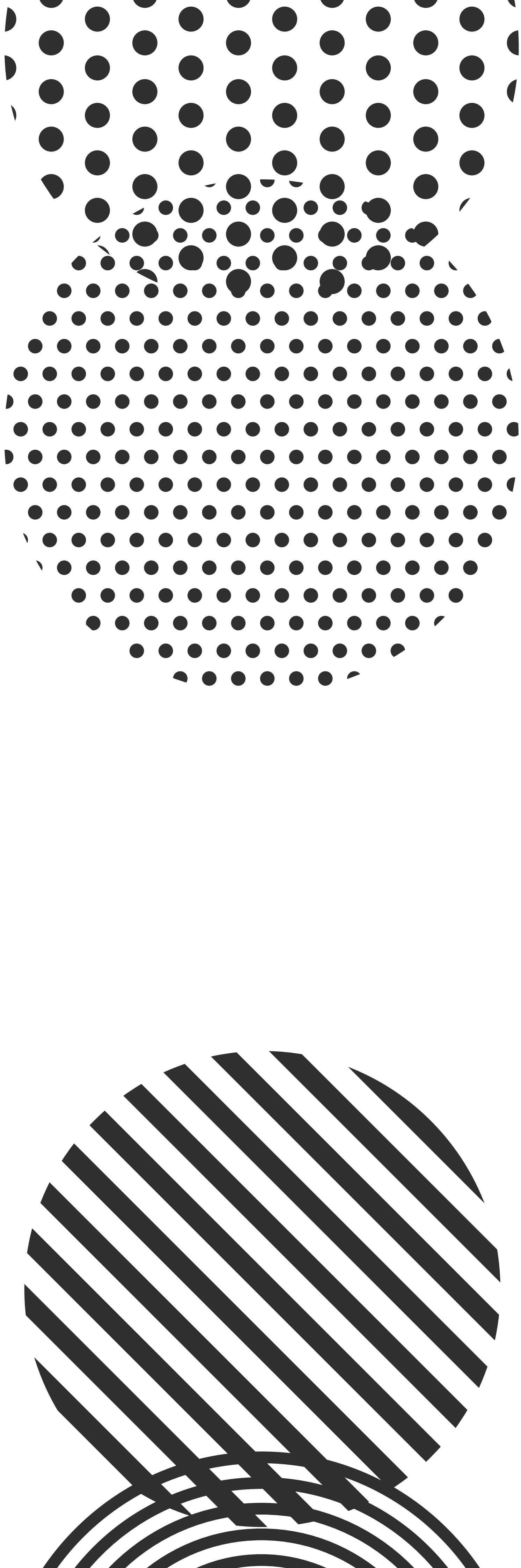

1 Show that the matrices $D(g)^T$ (in contrast to $D(g)^*$) do not form a representation.

We can do so by transposing $D(g_1)D(g_2) = D(g_1 g_2)$

$$\Rightarrow \quad [D(g_1)D(g_2)]^T = D^T(g_1 g_2)$$

$$D^T(g_2)\,D^T(g_1) = D^T(g_1 g_2)$$

$$\cancel{D^T(g_2 g_1) = D^T(g_1 g_2)}$$ which is not true for non abelian G.

$\Rightarrow$ $D^T(g)$ don't form a representation because $D^T(g_1)\,D^T(g_2) \neq D^T(g_1 g_2)$

2 Show that squares of elements that are equivalent to each other are also equivalent to each other.

$$g \sim g' \quad \Leftrightarrow \quad g = f^{-1} g' f$$

$$\Rightarrow \quad (g')^2 \quad \Rightarrow \quad f^{-1}(g')^2 f = f^{-1} g' f f^{-1} g' f = g \cdot g = g^2$$

$\Rightarrow$ $g^2 \sim g'$, i.e squares of elements equivalent to each other are also equivalent to each other.

3 Verify that the 2-dimensional irreducible representation of D_4, the invariance group of the square, is real, as is almost self-evident geometrically.

The character table part of 2-d rep of D_4 is

2	$n_c = 1$
-2	$n_c = 1$
0	$n_c = 2$
0	$n_c = 2$
0	$n_c = 2$

And we can use reality checker:

$$\sum_{g \in G} \chi^{(r)}(g^2) = \eta^{(r)} N(G)$$, where $$\eta^{(r)} = \begin{cases} 1, & \text{if real} \\ -1 & \text{if pseudo real} \\ 0, & \text{if complex} \end{cases}$$

$$\sum_{g\in G} \chi^{(r)}(g^2) = 2\cdot 1 + 2\cdot 1 - 2\cdot 2 + (2\cdot 2)\cdot 2$$
$$= 4 - 4 + 8 = 8 = N(G)$$

$I^2 = I$ $\quad n_c = 1$

$(-I)^2 = I$ $\quad n_c = 1$

$(R)^2 = -I$ $\quad n_c = 2$

$r_x^2 = I$ $\quad n_c = 2$

$d^2 = I$ $\quad n_c = 2$

$\Rightarrow$ D_4's 2-d rep is real.

We use $\chi(I) = 2$ & $\chi(-I) = -2$ & n_c's

4 Give the 3 of A_4 a reality check.

3 of A_4 is:

A_4	n_c	c	3	
	1	I	3	$I^2 = I$
	3	$(12)(34)$	-1	$((12)(34))^2 = I$
	4	(123)	0	$(123)(123) = (321) \to \chi(c) = 0$
	4	(134)	0	$(134)(134) = (431) \Rightarrow \chi(c) = 0$

Reality checker:

$$\sum_{g\in G} \chi^{(r)}(g^2) = \eta^{(r)} N(G)$$, where $\eta^{(r)} = \begin{cases} 1, & \text{if real} \\ -1 & \text{if pseudo real} \\ 0, & \text{if complex} \end{cases}$

$\Rightarrow$ $1\cdot \chi^{(c)}(I) + 3\cdot \chi^{(c)}(I) = 1\cdot 3 + 3\cdot 3 = 12$ $\quad // \; N(G) = \frac{4!}{2} = 12$

$= N(G) \Rightarrow$ Real

5 Give the 3 of S_4 a reality check.

S_4	n_c	c	3	c^2
	1	I	3	$I^2 = I \Rightarrow \chi(c) = 3$
	3	$(12)(34)$	-1	$((12)(34))^2 = I \Rightarrow \chi(c) = 3$
	8	(123)	0	$(123)(123) = (321) = (132)$ same class $\chi(c) = 0$
	6	(12)	1	$(12)(12) = I \Rightarrow \chi(c) = 3$
	6	(1234)	-1	$(1234)(1234) = (42)(13) = (13)(42) \Rightarrow \chi(c) = -1$

$$\Rightarrow \sum_{g \in G} \chi^{(r)}(g^2) = 1 \cdot 3 + 3 \cdot 3 + 8 \cdot 0 + 6 \cdot 3 + 6 \cdot (-1)$$

$$= 30 - 6 = 24 = N(G) \qquad // \; N(G) = 4! = 12 \cdot 2 = 24$$

$\Rightarrow$ Real

6 Show that the 2 of the quarternionic group is pseudoreal.

2 of quarternionic group:

Q	n_c	class	2 (character)	c^2
	1	1	2	$1^2 = 1^2 \Rightarrow \chi(c) = 2$
	1	-1	-2	$-1^2 = 1 \Rightarrow \chi(c) = 2$
	2	$i, -i$	0	$i^2 = -1 \to \chi(c) = -2$
	2	$j, -j$	0	$j^2 = -1 \to \chi(c) = -2$
	2	$k, -k$	0	$k^2 = -1 \to \chi(c) = -2$

$$\Rightarrow \sum_g \chi(g^2) = 1 \cdot 2 + 1 \cdot 2 + 2 \cdot (-2 - 2 - 2) \qquad // \; N(G)$$

$$= 4 - 12 = -8 = -N(G) \Rightarrow \text{Pseudo real}$$

7 How many square roots does (123) in A_4 have?

$$\sigma_{(123)} = \sum_r \eta^{(r)} \chi^{(r)}((123)) \qquad A_4 \text{ has two real representations: } 1, 3$$

$$\Rightarrow \sigma_{(123)} = \chi^{(1)}((123)) + \chi^{(3)}((123)) \qquad // \text{ From character table of } A_4 \text{ we get } \chi^{(1)}(123) = 1, \; \chi^{(3)}(123) = 0$$

$$\sigma_{(123)} = 1$$

i.e (123) only has one square root

8 For S_4, evaluate $\sigma_{(12)(34)}$, $\sigma_{(123)}$, $\sigma_{(12)}$, and $\sigma_{(1234)}$, and check your answers against the multiplication table.

S_4 has the following char table:

S_4	n_c		1	$\bar{1}$	2	3	$\bar{3}$
	1	I	1	1	2	3	3
Z_2	3	(12)(34)	1	1	2	−1	−1
Z_3	8	(123)	1	1	−1	0	0
Z_2	6	(12)	1	−1	0	1	−1
Z_4	6	(1234)	1	−1	0	−1	1

S_4 has only real representations:

- $\sigma_{(12)(34)} = \sum_r \eta^{(r)} \chi^{(r)}((12)(34)) = 1+1+2-1-1 = 2$

$\Rightarrow$ $(12)(34)$ has two square roots

- $\sigma_{(123)} = \sum_r \eta^{(r)} \chi^{(r)}((123)) = 1+1-1 = 1$

$\Rightarrow$ (123) has one square root

- $\sigma_{(12)} = \sum_r \eta^{(r)} \chi^{(r)}((12)) = 1-1+1-1 = 0$

$\Rightarrow$ (12) has zero square roots

- $\sigma_{(1234)} = \sum_r \eta^{(r)} \chi^{(r)}((1234)) = 1-1+1-1 = 0$

$\Rightarrow$ (1234) has zero square roots

Checked the ans in multiplication table & result same as it should be.

9 For A_5, evaluate σ_I, $\sigma_{(12)(34)}$, $\sigma_{(123)}$, $\sigma_{(12345)}$, and $\sigma_{(12354)}$, and check your answers against the multiplication table.

A_5 has character table:

We finally end up with

A_5	n_c		1	3	3	4	5	C^2
	1	I	1	3	3	4	5	$I^2 = I$
Z_2	15	$(12)(34)$	1	-1	-1	0	1	I
Z_3	20	(123)	1	0	0	1	-1	(132)
Z_5	12	(12345)	1	ζ	$1-\zeta$	-1	0	(13524)
Z_5	12	(12354)	1	$1-\zeta$	ζ	-1	0	(13425)

Same class

Let's check if these reps are real, pseudo or complex:

the trivial rep is real fine

$(12345)(12354)$

$(13)(24)5$

$$\Rightarrow \sum_g \chi^{(3)}(g^2) = 3 + 3\cdot 15 + 20\cdot 0 + 12\cdot\zeta + 12(1-\zeta)$$

$$= 3\cdot 16 + 12 = 3\cdot(16+4) = 3\cdot 20 = 60 = N(G)$$

$\Rightarrow$ Real (both 3's real)

$\| \; N(G) = \frac{5!}{2} = 5\cdot 4\cdot 3 = 60$

$$\sum_g \chi^{(4)}(g^2) = 4 + 4\cdot 15 + 1\cdot 20 - 1\cdot 12 - 1\cdot 12$$
$$= 84 - 24 = 60 = N(G) \Rightarrow \text{Real}$$

$$\sum_g \chi^{(5)}(g^2) = 5\cdot 1 + 15\cdot 5 - 20 = 15\cdot 4 = 60 = N(G) \Rightarrow \text{Real}$$

$$\Rightarrow \sigma_I = \sum_r d_{r_{\text{real}}} - \sum_r d_{r_{\text{pseud}}} = \sum_r d_{r_{\text{real}}} = 1 + 3\cdot 2 + 4 + 5 = 16$$

$\Rightarrow$ Identity has 16 square roots.

$$\sigma_{(12)(34)} = \sum_r \eta^{(r)} \chi^{(r)}((12)(34)) = \sum_r \chi^{(r)}((12)(34)) = 1 - 1 - 1 + 0 + 1 = 0$$

$\Rightarrow$ No square roots for $(12)(34)$

$$\sigma_{(123)} = \sum_r \chi^{(r)}((123)) = 1 + 1 - 1 = 1$$

$\Rightarrow$ one square root for (123)

$$\sigma_{(12345)} = \sum_r \chi^{(r)}((12345)) = 2 - 1 = 1 \Rightarrow \text{one squareroot for } (12345)$$

$$\sigma_{(12354)} = \sum_r \chi^{(r)}((12354)) = 2 - 1 = 1 \Rightarrow \text{one square root for } (12354)$$

10 Verify the (13) for $\sum_g D^{(r)}(g^2)$ in A_4.

$$\sum_g D^{(r)}(g^2) = N(G)(\eta^{(r)}/d_r)I \qquad (13)$$

Character table of A_4

A_4	n_c		1	1′	1″	3
	1	I	1	1	1	3
Z_2	3	(12)(34)	1	1	1	-1
Z_3	4	(123)	1	ω	ω^*	0
Z_3	4	(132)	1	ω^*	ω	0

$1'$ & $1''$ are manifestly complex

$(123)(123) = (132)$
$(132)(132) = (123)$

$$\Rightarrow \sum_g \chi^{(1'')}(g^2) = 4(1 + \omega + \omega^*) = 0$$

for both $1'$ & $1''$

trivial rep is real & for 3: $\sum_g \chi^{(1)}(g^2) = 3 + 3\cdot(-1) = 0$

$\Rightarrow$ Complex

$$\sum_g D^{(1)}(g^2) = 1 + 3\cdot 1 + 4\cdot 1 + 4\cdot 1 = 12 = N(G) = \frac{N(G)}{d_r}\eta^{(r)} I$$

Since for triv. rep. $\eta^{(r)} = 1 = d_r$ & $I = \mathbb{1}$

For $1'$ & $1''$ $\sum_g D^{(1')}(g^2) = 1 + 3 + 4\omega + 4\omega^* = 4(1+\omega+\omega^*) = 0$

$$= \frac{N(G)}{d_r}\eta^{(r)} I \quad \text{since } \eta^{(r)} = 0$$

For 3:

$$\sum_g D^{(1'')}(g^2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + 3\cdot\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix} + \underset{(123)}{\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}} + \begin{pmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} + \dots \text{(2 more)}$$

$$+ \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} + \dots \text{(3 more)} = 0$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

11 For A_4, verify the result for the number of solutions of $f^2 g^2 = I$.

The char table of A_4 is:

A_4	n_c		1	$1'$	$1''$	3
	1	I	1	1	1	3
Z_2	3	$(12)(34)$	1	1	1	-1
Z_3	4	(123)	1	ω	ω^*	0
Z_3	4	(132)	1	ω^*	ω	0
		$\eta^{(r)}$	1	0	0	1

$\Rightarrow N(G) = 12$

For identity:

$$\tau_I = N(G) \sum_r (\eta^{(r)})^2$$

$\Rightarrow \tau_I = 12 \cdot \sum_r (\eta^{(r)})^2$

$= 12 \cdot (1^2 + 1^2) = 24$

$I^2 I^2 = I$, $\quad ((12)(34))^2 = I$